

Homework (Jalapeno)

Q1. Solve the quadratic equation: $x^2 + 5x + 6 = 0$

- (A) $x = -2, -3$
- (B) $x = -1, -6$
- (C) $x = -2, -4$
- (D) $x = 1, -6$
- (E) $x = 3, -2$

Q2. Factor the quadratic expression: $x^2 + 4x + 4$

- (A) $(x + 2)^2$
- (B) $(x + 1)^2$
- (C) $(x + 3)^2$
- (D) $(x - 2)^2$
- (E) $(x - 1)^2$

Q3. The ancient Egyptians built a rectangular pond with a length of $x + 3$ and a width of $x - 2$. If the area is 15, solve for x .

- (A) $x = 5$
- (B) $x = 7$
- (C) $x = 3$
- (D) $x = 2$
- (E) $x = 1$

Q4. The quadratic equation $x^2 - 7x + 12 = 0$ has solutions

- (A) $x = 3, 4$
- (B) $x = 2, 6$
- (C) $x = 1, 12$
- (D) $x = 5, 12$
- (E) $x = 3, 12$

Q5. Factor the quadratic expression: $x^2 - 9x + 20$

- (A) $(x - 5)(x - 4)$
- (B) $(x - 4)(x - 5)$
- (C) $(x + 4)(x + 5)$
- (D) $(x + 5)(x - 4)$
- (E) $(x - 4)(x + 5)$

Q6. Solve the quadratic equation: $x^2 - 2x - 15 = 0$

- (A) $x = -3, 5$
- (B) $x = 3, -5$
- (C) $x = 1, 15$
- (D) $x = -1, -15$
- (E) $x = 2, 15$

Q7. The ancient Greeks used quadratic equations to design their stadiums. If a stadium has a length of $x + 2$ and a width of $x - 1$, and the area is 20, solve for x .

- (A) $x = 5$
- (B) $x = 4$
- (C) $x = 6$
- (D) $x = 3$
- (E) $x = 2$

Q8. Factor the quadratic expression: $x^2 + 2x - 15$

- (A) $(x - 5)(x + 3)$
- (B) $(x + 5)(x - 3)$
- (C) $(x - 3)(x + 5)$
- (D) $(x + 3)(x - 5)$
- (E) $(x + 1)(x - 15)$

Open Ended Questions (FRQ)

Q9. The ancient Babylonians used quadratic equations to calculate the area of their fields. If a field has a length of $x + 1$ and a width of $x - 2$, and the area is 24, solve for x and show your work.

Q10. The quadratic equation $x^2 + 5x + 6 = 0$ has solutions. Verify the solutions and show your work.

Chef's Tips

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- Tip 1: Encourage students to use factoring to solve quadratic equations.
- Tip 2: Remind students to verify their solutions.
- Tip 3: Provide feedback on student work to help them improve.